[**REC:** Title is strong, accurate, and concise. An alternative emphasizing deterministic chaos suppression could be: "Synchronization, Chaos Suppression, and Spatiotemporal Phases in Driven Precessing Domain Walls".]
[**REC:** PACS / PhysSH Codes: Replace placeholder with valid PRB codes: `75.78.Fg` (Dynamics of domain structures), `75.60.Ch` (Domain walls and domain structure), `05.45.Xt` (Synchronization; coupled oscillators), `05.45.Pq` (Numerical simulations of chaotic systems).]

Synchronization and Spatiotemporal Phases in Driven Precessing Domain Walls

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The dynamics of extended magnetic domain walls driven by constant and alternating magnetic fields are studied using a collective coordinate u–ϕ model. While purely constant-field driven systems exhibit deterministic spatiotemporal chaos and roughening, the alternating component can lock the precessional frequency via Shapiro-like steps. Using linear stability analysis and Floquet theory, the phase boundaries of stable, synchronized flat wall states are mapped. By studying the nonlinear dynamics of the extended system, a rich phase diagram is identified, composed of five dynamical regimes characterized by the combination of domain wall spatial order (flat, rough or periodic) and domain wall precession temporal order (pinned, synchronized or unsynchronized). Notably, the macroscopic temporal order of the synchronized phases can lead to a flat domain wall or to a periodic structure dominated by a narrow band of linearly unstable modes, both with coherent precession of the domain wall magnetization, while unsynchronized phases can be flat or rough structures with spatiotemporal chaos and incoherent precession. [**REC:** Abstract suggestion: Conclude the abstract by explicitly mentioning the dynamic universality class of the rough phase: "We demonstrate that the non-synchronized rough phase belongs to the one-dimensional Edwards–Wilkinson universality class with roughness exponent $\zeta = 1/2$ and dynamic exponent $z = 2$. In the synchronized periodic phase, phase-locking arrests the turbulent mode cascade, selecting an intrinsic spatial wavelength $\kappa^* \approx 0.84$ that persists in the thermodynamic limit."]

I. INTRODUCTION

Synchronization is a generic phenomenon in nonlinear dynamical systems whereby oscillators adjust their rhythms through mutual interactions or external periodic forcing. It occurs in a wide variety of physical, biological and technological systems, including Josephson junctions, coupled pendulums and neuronal networks[1,2].

Magnetic domain walls (DWs) can also behave like oscillators. When driven by a field higher than the Walker breakdown field[3], the internal magnetization angle begins to rotate continuously at a specific angular velocity. Within the framework of the 1D Walker model, adding a harmonic component to the driving field produces Shapiro steps[4–6]: the precessional frequency of the internal degree of freedom remains fixed to an integer multiple of the driving frequency over a range of parameters.

While insightful, the 1D Walker model fails to capture the effects of the DW flexural modes[7–9]. In a more realistic model for an extended DW, it has been recently shown[10] that spatiotemporal chaos can emerge when the wall enters the precessional regime. Even in this complex condition, the corresponding rigid wall frequency still appears in the power spectrum above the chaotic broadband baseline noise.

This naturally raises the question of whether synchrony can occur in extended DWs. In that case, the macroscopic order induced by the synchrony should manifest itself in the spatial structure of the DW. Similar effects have been found in simulations[11,12] and experimentally[13] in other physical systems susceptible to synchrony. An alternating magnetic field or electrical current via spin transfer torque could thus serve to control the otherwise complex DW dynamics, a capability highly relevant for DW-based technologies[6,14–16].

In this work, we study the dynamics of extended DWs with perpendicular magnetic anisotropy using the collective-coordinate $u - \phi$ model[3,17]. We analyze the effects of an AC driving field on the spatiotemporal dynamics of the DW over a wide range of system sizes. In particular, we demonstrate that an appropriate AC field can suppress the spatiotemporal chaos of the DC-driven wall. We identify the parameter regions in which the DW synchronizes with the external drive and determine how synchronization affects its long-range spatial structure. This reveals a phase diagram with five distinct dynamical regimes, characterized by the combination of the DW spatial structure (flat, rough or periodic) and the temporal dynamics of the DW precession (pinned, synchronized or unsynchronized). In the thermodynamic limit, these regimes are separated by nonequilibrium dynamical phase transitions. [**REC:** Paper Outline: Conclude the Introduction with an explicit roadmap paragraph: "The remainder of this article is organized as follows. Section III describes the u–ϕ model equations and presents the first- and second-order perturbation theory for rigid

DW synchrony. Section IV formulates the linear stability problem via Floquet theory and maps the instability bands of synchronized flat walls. Section V examines the nonlinear dynamics of extended walls under sweeps of the DC driving field. Section VI constructs the full dynamical phase diagram, characterizes the dynamic Edwards–Wilkinson scaling of the rough phase, and analyzes the finite-size spatial ordering of the periodic phase. Concluding remarks are provided in Section VII, while appendices detail experimental parameters, numerical schemes, small-system finite-size effects, second-order calculations, and multimode secular expansions."]

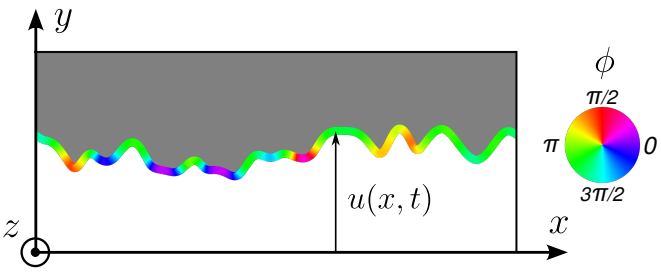


Figure 1. Snapshot of a DW configuration highlighting its collective coordinates $u(x,t)$ and $\phi(x,t)$. Reproduced from Ref. 10.

II. MODELS AND METHODS

Our fundamental description of the DW dynamics is based on the Landau-Lifshitz-Gilbert (LLG) equation:

$$\frac{\partial \mathbf{m}}{\partial t} = -\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}} + \alpha \mathbf{m} \times \frac{\partial \mathbf{m}}{\partial t}\,, \tag{1}$$

where γ is the reduced gyromagnetic ratio and α is the Gilbert damping constant. The unit magnetization vector is defined in spherical coordinates as $\mathbf{m} = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$. The effective field $\mathbf{H}_{\text{eff}} = -(1/\mu_0 M_s)\delta E/\delta \mathbf{m}$ is derived from the total magnetic energy functional, which incorporates exchange, anisotropy, Zeeman, and magnetostatic contributions:

$$E = \int \Big\{ A_{\text{ex}}[(\nabla\theta)^2 + \sin^2\theta(\nabla\varphi)^2] - K_u\cos^2\theta \\ -\mu_0 H_z(t) M_s \cos\theta + \frac{1}{2}\mu_0 M_s^2 [N_n(\mathbf{m}\cdot\mathbf{n})^2 + \cos^2\theta] \Big\} d\mathbf{r}\,. \tag{2}$$

Here M_s is the saturation magnetization, A_{ex} is the exchange stiffness constant, K_u the uniaxial anisotropy constant and μ_0 the magnetic permeability. We consider a uniform, time-dependent magnetic field $H_z(t)$ along the z-axis. The last contribution in Eq. (2) accounts for magnetostatic effects of a DW in the local thin-film approximation, where N_n is the demagnetizing factor along the DW normal direction[18]. We assume a directed DW with its average normal in the direction $\mathbf{n} \equiv \hat{\mathbf{y}}$.

To gain additional insight, we describe the DW dynamics using a reduced collective-coordinate framework known as the $u-\phi$ model. This approach captures the most relevant degrees of freedom while remaining computationally efficient compared to full micromagnetic simulations of the LLG equation.

[**REC:** Figure 1 caption enhancement: Expand caption to explain the visual elements: "Snapshot of an extended DW configuration displaying the spatial coordinate x along the wall, the transverse position $u(x,t)$ (solid curve), and the local in-plane magnetization angle $\phi(x,t)$ (arrows and color gradient). Local twists in ϕ where the angle rotates by $\pm\pi$ correspond to vertical Bloch lines separating Néel-like segments."]

Substituting a Bloch domain-wall ansatz

$$\theta(x,y,t) = 2\arctan\{\exp[(y-u(x,t))/\Delta]\}\,, \tag{3}$$
$$\varphi(x,y,t) = \phi(x,t), \tag{4}$$

into Eq. (1), gives us the reduced coordinate model for the DW. Here $u(x,t)$ and $\phi(x,t)$ represent the local position and the internal magnetization angle of the wall, respectively (see Fig. 1). We assume a constant DW width $\Delta = \sqrt{A_{\text{ex}}/(K_u - \mu_0 M_s^2/2)}$, where the denominator accounts for the effective anisotropy including the magnetostatic demagnetizing energy.

This model treats the DW as a single-valued directed interface, thereby prohibiting overhangs and loops while allowing for the proliferation and dynamics of vertical Bloch lines. It therefore goes beyond standard elastic interface models by incorporating an internal degree of freedom. We do not consider here the additional effects of thermal fluctuations and quenched disorder.

To simplify the analysis, we nondimensionalize the model using characteristic length, time, and magnetic-field scales[10]. Distances are measured in units of $u_0 = \Delta$ along the y-direction, and $L_0 = \Delta(2\tilde{K}/N_n)^{1/2}$ in the x-direction, where $\tilde{K} = 2K_u/(\mu_0 M_s^2) - 1$. Time is measured in units of $T_0 = (1+\alpha^2)/(\gamma M_s N_n)$, and the applied field $H_z(t)$ is expressed in units of the Walker breakdown field[18] $H_W = \alpha M_s N_n/2$, with $h(t) = H_z(t)/H_W$. See Appendix A for details on the conversion to physical units for a thin Co/Pt multilayer. The resulting equations constitute the u–ϕ or Slonczewski model[17]:

$$\dot{u} = \frac{1}{2}\left[\alpha^2 h(t) + \sin(2\phi) + \alpha\partial_x^2 u - \partial_x^2\phi\right], \\ \dot{\phi} = \frac{1}{2}\left[\alpha h(t) - \alpha\sin(2\phi) + \partial_x^2 u + \alpha\partial_x^2\phi\right]. \tag{5}$$

This model reduces to the Walker model[3] for rigid DWs, where $\partial_x^2 u = \partial_x^2\phi = 0$. To study synchronization phenomena in extended DWs we set $h(t) = h_0 + h_{\text{a}}\cos(\Omega t)$, with h_0 the DC field and h_{a} the AC field amplitude. Note that the unit of frequency in this system is $1/T_0$. We then numerically solve the full system of coupled, non-autonomous nonlinear partial differential equations using a pseudo-spectral Crank–Nicolson scheme (see Appendix B for details).

III. SYNCHRONY IN RIGID DOMAIN WALLS

Given that the extended DW can become flat under certain conditions, it is relevant to study synchrony in the resulting rigid DW, described by the Walker model. This rigid-DW regime represents the simplest form of macroscopic order the DW can exhibit, and is considerably more analytically tractable than the full extended model.

To simplify the analysis, we define the variables $\tau \equiv \Omega t$, $q(x,\tau) \equiv u(x,\tau/\Omega)$ and $\psi(x,\tau) \equiv 2\phi(x,\tau/\Omega)$. If we assume a spatially uniform DW given by $q(x,\tau) = q_1(\tau)$ and $\psi(x,\tau) = \psi_1(\tau)$, Eqs. (5) yields the following system of coupled ordinary differential equations for q_1 and ψ_1:

$$\dot{q}_1 = \frac{1}{2\Omega}\left[\alpha^2 h_0 + \alpha^2 h_\mathrm{a}\cos(\tau) + \sin(\psi_1)\right], \tag{6}$$

$$\dot{\psi}_1 = \frac{\alpha}{\Omega}\left[h_0 + h_\mathrm{a}\cos(\tau) - \sin(\psi_1)\right], \tag{7}$$

The period of the non-autonomous term is fixed to 2π, and Eq. (7) depends only on the parameters h_0, h_a and the ratio α/Ω. We will focus on this equation since it is decoupled from the q_1 solution and dictates the overall dynamics of the system.

For the case $h_\mathrm{a} = 0$, Eq. (7) reduces to the Adler equation[19], for which there is an analytical expression for the time average $\langle\dot{\psi}_1\rangle_\tau$:

$$\langle\dot{\psi}_1\rangle_\tau = \begin{cases} 0 & \text{if } |h_0| \le 1, \\ \mathrm{sgn}(h_0)\dfrac{\alpha}{\Omega}\sqrt{h_0^2 - 1} & \text{if } |h_0| > 1. \end{cases} \tag{8}$$

In addition, the average velocity of q_1 is given by

$$\langle\dot{q}_1\rangle_\tau = \frac{\alpha^2+1}{2\Omega}h_0 - \frac{\alpha}{2}\langle\dot{\psi}_1\rangle_\tau. \tag{9}$$

Note that this expression is also true for the center of mass of the flexible wall model with periodic boundary conditions. Therefore, for the time-independent driving case we obtain

$$\langle\dot{q}_1\rangle_\tau = \begin{cases} \dfrac{\alpha^2+1}{2\Omega}h_0 & \text{if } |h_0| \le 1\ , \\ \dfrac{\alpha^2+1}{2\Omega}h_0 - \dfrac{\alpha^2}{2\Omega}\,\mathrm{sgn}(h_0)\sqrt{h_0^2-1} & \text{if } |h_0| > 1\ . \end{cases} \tag{10}$$

When $h_\mathrm{a} \neq 0$, solutions must be found numerically (see Appendix B). To determine if a stable periodic solution exists, we recorded the ψ_1 variable modulo 2π at every period of the drive: the system has converged to a stable periodic solution if and only if these values converge to a single equilibrium value ψ_eq, independent of the initial condition.

Synchrony is expected to occur near the values of h_0 for which Eq. (8) yields an integer. Figure 2 shows example curves of the steady-state average velocities of both q_1 and ψ_1 as a function of the DC field h_0, with the white regions marking the synchronization steps where $\langle\dot{\psi}_1\rangle_\tau = n$, and $n = 0, 1, 2$. A Walker-breakdown-like feature repeats in $\langle\dot{q}_1\rangle_\tau$ at each frequency-unlocking transition of the internal degree of freedom ψ_1. Within each synchronization step, the translational average velocity is

$$\langle\dot{q}_1\rangle_\tau = \frac{\alpha^2+1}{2\Omega}h_0 - \frac{\alpha}{2}n, \quad h_1^n \le h_0 \le h_2^n, \tag{11}$$

with $h_1^n < h_2^n$ delimiting the extension of the n-th step. Consequently, the DW mobility within a synchronized state is identical to that of the constant-driving-field case below the Walker field ($|h_0| < 1$).

These synchronization regions in $\langle\dot{\psi}_1\rangle_\tau$ are known as *Shapiro steps*, as they are analogous to those predicted and observed in the I–V curves of driven overdamped Josephson junctions[4]. [**REC:** Figure 2 suggestions: (i) In the caption, explicitly label panel (a) top and (b) bottom: "(a) Steady-state average angular velocity $\langle\dot{\psi}_1\rangle_\tau$ and (b) translational velocity $\langle\dot{q}_1\rangle_\tau$..." (ii) In the text, note that within each Shapiro step, the differential mobility $d\langle\dot{q}_1\rangle_\tau/dh_0 = (\alpha^2+1)/(2\Omega)$ is strictly constant and coincides with the linear sub-Walker mobility, whereas outside the steps the velocity collapses due to chaotic/irregular precession."]

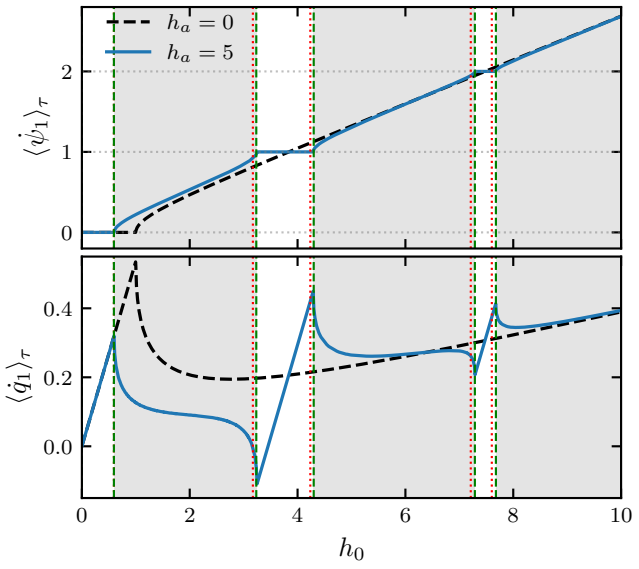


Figure 2. Steady-state average velocities of the internal (ψ_1, top) and translational (q_1, bottom) degrees of freedom as a function of the DC field h_0, for $h_a = 0$ (dashed black) and $h_a = 5$ (solid blue). White regions indicate the synchronization steps, where $\langle\dot{\psi}_1\rangle_\tau$ locks to integer multiples of the driving frequency (dotted horizontal lines, top panel). Vertical red dotted lines indicate synchrony boundaries as given by Eq. (24); their second-order approximation, given by Eq. (D9), is shown as green dashed lines. Walker-breakdown-like spikes in $\langle\dot{q}_1\rangle_\tau$ recur at each unlocking transition of ψ_1.

In a synchronized state, solutions should be of the form

$$\psi_\mathrm{s}(\tau) = \psi_\mathrm{c} + n\tau + \tilde{\psi}(\tau), \tag{12}$$

with ψ_c a constant, $\tilde{\psi}(\tau)$ a 2π-periodic function with zero mean, and n an integer. In this case, it is clear that $\langle \dot{\psi}_s \rangle_\tau = n$.

To approximate the boundaries of the parameter regions for which synchronized solutions exist, we can treat the nonlinear $\sin\psi_1$ term as a perturbation by defining the scaled variables $\tilde{h}_0 \equiv \epsilon h_0$, $\tilde{h}_a \equiv \epsilon h_a$, and introducing $\epsilon \equiv \alpha/\Omega$ as an explicit small parameter. Equation (7) then reads:

$$\dot{\psi}_1 = \tilde{h}_0 + \tilde{h}_a \cos(\tau) - \epsilon \sin(\psi_1). \tag{13}$$

We now expand both the solution ψ_1 and the DC driving parameter $\tilde{h}_0$ in power series of ϵ:

$$\psi_1(\tau) = \psi^{(0)}(\tau) + \epsilon\psi^{(1)}(\tau) + \epsilon^2\psi^{(2)}(\tau) + \dots, \tag{14}$$

$$\tilde{h}_0 = \tilde{h}_0^{(0)} + \epsilon\tilde{h}_0^{(1)} + \epsilon^2\tilde{h}_0^{(2)} + \dots \tag{15}$$

Substituting these expansions into Eq. (13) and collecting terms of order ϵ^0, we obtain the unperturbed equation:

$$\dot{\psi}^{(0)} = \tilde{h}_0^{(0)} + \tilde{h}_a \cos(\tau). \tag{16}$$

Phase-locking to the n-th step requires $\langle \dot{\psi}^{(0)} \rangle_\tau = n$, which fixes $\tilde{h}_0^{(0)} = n$. Integrating yields the zeroth-order solution:

$$\psi^{(0)}(\tau) = n\tau + \tilde{h}_a \sin(\tau) + \psi_c. \tag{17}$$

At order ϵ^1, the dynamics of the first-order correction $\psi^{(1)}$ are governed by:

$$\dot{\psi}^{(1)} = \tilde{h}_0^{(1)} - \sin\left(\psi^{(0)}\right). \tag{18}$$

For the synchronized solution to remain valid at this order, $\psi^{(1)}$ must remain a 2π-periodic function of zero mean. Therefore, the time average of the right-hand side of Eq. (18) must vanish:

$$\tilde{h}_0^{(1)} = \left\langle \sin\left(\psi^{(0)}\right) \right\rangle_\tau. \tag{19}$$

Expanding $\sin\left(\psi^{(0)}\right)$ using the Jacobi–Anger identity,

$$\sin\left(\psi^{(0)}\right) = \sum_{m=-\infty}^{\infty} J_m(\tilde{h}_a) \sin[(n+m)\tau + \psi_c], \tag{20}$$

we see that every term oscillates with zero mean except the one with $m = -n$. This DC component yields:

$$\tilde{h}_0^{(1)} = J_{-n}(\tilde{h}_a)\sin(\psi_c) = (-1)^n J_n(\tilde{h}_a)\sin(\psi_c). \tag{21}$$

Truncating the expansion (15) at first order gives $\tilde{h}_0 \approx n + \epsilon\tilde{h}_0^{(1)}$. Substituting back the original physical parameters h_0, h_a, α/Ω, and the solution (21) for $\tilde{h}_0^{(1)}$, we obtain:

$$h_0 \approx \frac{n\Omega}{\alpha} + (-1)^n J_n\left(\frac{\alpha}{\Omega}h_a\right)\sin(\psi_c). \tag{22}$$

Since $\sin\psi_c \in [-1, 1]$, a synchronized solution exists only if

$$\left| h_0 - \frac{n\Omega}{\alpha} \right| \le \left| J_n\left(\frac{\alpha}{\Omega}h_a\right) \right|. \tag{23}$$

Finally, each step is centered at $h_0 = n\Omega/\alpha$ and has a field-range width

$$h_2^n - h_1^n = 2\left| J_n\left(\frac{\alpha}{\Omega}h_a\right) \right|. \tag{24}$$

The details of the continuation of this perturbative treatment to second order in ϵ are worked out in Appendix D. The only second order effect is a rigid shift depending only on $\alpha h_a/\Omega$ in the synchronization steps with $|n| > 0$. The correction is more significant for the smaller absolute values of n and h_a, vanishing in the limits where either variable goes to infinity. For the $n = 0$ step, the second order corrections vanish identically. In Fig. 2 we can compare first and second order corrections. [**REC:** Second-order perturbation physical interpretation: Emphasize that the second-order shift $\Delta h_0^{(2)} = \frac{\alpha}{\Omega}A$ arises from the asymmetric time spent by the magnetization vector traversing the attractive versus repulsive quadrants of the nonlinear $\sin(\psi_1)$ torque. Because $A \approx \frac{1}{2n}$ as $h_a \to 0$, this shift is always toward higher DC fields, explaining why numerical Arnold tongues tilt slightly toward larger driving fields for finite damping α.]

IV. LINEAR STABILITY OF FLAT, SYNCHRONIZED DOMAIN WALLS

To study the linear stability of synchronized flat solutions, we adopt the theoretical framework recently detailed by Pusiol et al.[10] and extend it to account for a time-dependent magnetic field. We linearize a solution of Eq. (5) $\mathbf{v}(x,\tau) \equiv (u, \phi)^T$ around the flat-wall solution,

$$\mathbf{v}(x,\tau) = \mathbf{v}_1(\tau) + \boldsymbol{\eta}(x,\tau), \quad \boldsymbol{\eta}(x,t) \equiv \begin{pmatrix} Q(x,t) \\ \Phi(x,t) \end{pmatrix} \tag{25}$$

where $\mathbf{v}_1(\tau) \equiv (q_1(\tau), \psi_1(\tau)/2)^T$. Substituting into Eq. (5), the linearized system reads

$$\partial_\tau \boldsymbol{\eta} = \frac{1}{2\Omega}\,\mathcal{M}(\tau)\,\boldsymbol{\eta}, \tag{26}$$

with

$$\mathcal{M}(\tau) \equiv \begin{pmatrix} \alpha\,\partial_x^2 & -\partial_x^2 + 2\cos\psi_1(\tau) \\ \partial_x^2 & \alpha\,\partial_x^2 - 2\alpha\cos\psi_1(\tau) \end{pmatrix}. \tag{27}$$

We perform a Fourier transform in space, so that $\partial_x^2 \to -\kappa^2$. Since Eq. (26) is linear with x-independent coefficients, distinct modes κ decouple, yielding

$$\partial_\tau \boldsymbol{\eta}_\kappa = \frac{1}{2\Omega}\,\mathcal{M}_\kappa(\tau)\,\boldsymbol{\eta}_\kappa, \tag{28}$$

with

$$\mathcal{M}_\kappa(\tau) \equiv -\kappa^2 \begin{pmatrix} \alpha & -1 \\ 1 & \alpha \end{pmatrix} + 2\cos\psi_1(\tau) \begin{pmatrix} 0 & 1 \\ 0 & -\alpha \end{pmatrix}, \quad (29)$$

where the first term is the τ-independent diffusive coupling between Q_κ and Φ_κ, and the second is the 2π-periodic modulation set by the rigid-wall background solution $\psi_1(\tau)$. The stability of a linear system with periodic coefficients can be determined using Floquet theory. By Floquet's theorem, a general solution of Eq. (28) can be written as

$$\boldsymbol{\eta}_\kappa(\tau) = P_\kappa(\tau)\, e^{R_\kappa \tau}\, \boldsymbol{\eta}_\kappa(0), \quad (30)$$

where $P_\kappa(\tau)$ is a 2π-periodic matrix and R_κ is a constant matrix. The monodromy matrix $M_\kappa \equiv P_\kappa(2\pi)\, e^{2\pi R_\kappa}$ is obtained by integrating Eq. (28) over one period, using the canonical initial conditions $\boldsymbol{\eta}_\kappa(0) = (1,0)^T$ and $(0,1)^T$. The amplitude of the perturbation grows exponentially – indicating that the system is unstable – whenever the Floquet multiplier (eigenvalue of M_κ) of largest modulus, μ_κ, satisfies $|\mu_\kappa| > 1$. Since the modes κ are decoupled, their Floquet multipliers can be computed independently, allowing straightforward parallelization across κ. Furthermore, the rescaled time conveniently fixes the integration interval to 2π regardless of parameter values. [**REC:** Connection to Growth Rates and Hill Expansion: State explicitly that the continuous mode growth rate is $\lambda_\kappa = \frac{\Omega}{2\pi} \ln|\mu_\kappa|$. Modes with $\mathrm{Re}(\lambda_\kappa) > 0$ grow exponentially. This directly connects the numerical monodromy calculation to the analytical multi-harmonic Hill matrix expansion detailed in the Supporting Information (see Appendix E), where the eigenvalues of the secular matrix $\mathcal{H}(\kappa)$ yield λ_κ algebraically.]

Figure 3 shows the resulting stability diagrams obtained numerically for $h_a = 2$ (a) and $h_a = 3$ (b), at $\alpha = 0.27$ and $\Omega = 1$, as an illustrative example. Although the rigid DW is synchronized within the corresponding Shapiro steps, finite intervals of h_0 exhibit spatial instabilities. These unstable regions are generally located near the boundaries of the synchronization steps and are characterized by a finite range of wavenumbers with Floquet multipliers satisfying $|\mu_\kappa| > 1$. Thus, synchronization of the internal precessional dynamics if the rigid DW does not necessarily imply spatial stability of the extended flexible DW. Instead, a synchronized flat DW can coexist with a band of unstable spatial modes, whose growth rates depend continuously on κ. The nonlinear evolution of these instabilities, and also the possible instabilities of non synchronized flat walls (for which Floquet analysis is not adapted), and their effect on the long-time DW structure are investigated in the following section. [**REC:** Figure 3 Discussion & Typo: (i) Fix typo: "internal precessional dynamics if the rigid DW" → "internal precessional dynamics of the rigid DW". (ii) Physical explanation of Fig. 3 features: Highlight that the instability bands (colored tongues) are parametric resonances occurring near $\kappa^2 \approx m\Omega$. For $h_a = 2$ (weak drive), individual tongues corresponding to $m = 0, 1, 2$ are well-separated. For $h_a = 3$ (stronger drive), inter-channel coupling hybridizes adjacent resonances, opening a reentrant stable region (a "waist") where the flat synchronized wall is restored. This non-monotonic stability window directly determines the morphology of the dynamical phase diagram.]

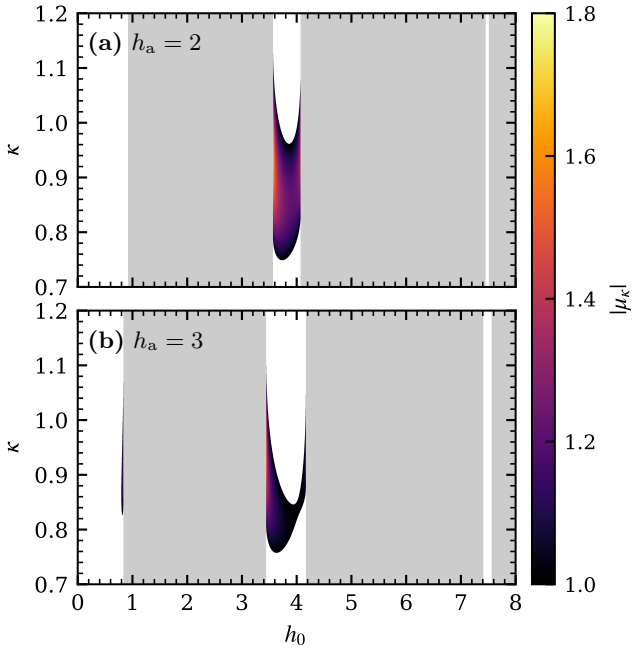


Figure 3. Linear stability diagram of DW modes for $\alpha = 0.27$ and $\Omega = 1$, with $h_a = 2$ in (a) and $h_a = 3$ in (b). White regions are stable; gray regions do not admit a stable time-periodic solution for the rigid wall. Color indicates the magnitude of the largest Floquet multiplier for unstable regions with $|\mu_\kappa| > 1$.

V. NONLINEAR DYNAMICS OF EXTENDED DWS

We now investigate the long-time dynamics under the combined DC+AC field, going beyond the linear stability analysis of rigid synchronized states. As an illustrative example, we numerically simulate the system of Eqs. (5) (see Appendix B) for the same parameters of the stability map of Fig. 3(b). We characterize the resulting dynamics through the mean angular velocity $\langle\dot{\phi}\rangle_{x,t}$, the mean DW velocity $\langle\dot{u}\rangle_{x,t}$, and the roughness

$$\langle w^2\rangle_t = \left\langle \langle u^2\rangle_x - \langle u\rangle_x^2 \right\rangle_t . \quad (31)$$

The time averages are taken after the system has reached its long-time dynamical regime.

We characterize synchronization with the external drive through the quantity $\langle\sigma^2\rangle_x$ by sampling the dis-

placement of the DW $u(x, t + 2T_{\rm drive}) - u(x, t)$ stroboscopically, at intervals of twice the period of the drive $T_{\rm drive} = 2\pi/\Omega$. For each point on the wall, we compute the variance of these displacements over time, and then we average the variances across the wall. When the system is phase-locked to the drive, the stroboscopic displacement at each point is identical in every cycle, resulting in $\langle\sigma^2\rangle_x \approx 0$. Otherwise, the displacement fluctuates over successive cycles, yielding a finite value of $\langle\sigma^2\rangle_x$. Sampling at twice the driving period allows us to detect synchrony to both harmonic and subharmonic frequencies.

To highlight finite-size effects, we consider a small system with $L/L_0 = 8$, for which the first nonzero mode is $\kappa = 2\pi/8 \approx 0.785$, and a large system with $L/L_0 = 1000$. The latter provides access to the extended-wall regime in which deterministic spatiotemporal chaos occurs under purely DC driving[10], whereas the small system is strongly affected by the discrete set of available spatial modes. This comparison allows us to distinguish finite-size effects from the behavior associated with a broad spectrum of flexural modes. [**REC:** System Size Accuracy in Text: Change "$L/L_0 = 1000$" to "$L/L_0 = 1024$" to match Figure 4 and its caption where the simulated large system is explicitly $L/L_0 = 1024$.]

Sacaria la fig 4(a), ya que podriamos correlacionar todo con la Fig 3(b) pero si ponemos $h_a = 3$ en vez de 4 y con rango extendido $0 < h < 8$.

- Describir las tres figuras cualitativamente.
- Describir zonas donde tenemos Floquet.
- Describir zonas donde no tenemos Floquet, pero sin embargo puede o no haber estabilidad.
- Concluir con la existencia de varias fases para dar pie a la proxima section.

[**REC:** Section V Text Reconstruction: The Spanish bullet points left by Ale (lines 811–816 above) and Luz's note on the synchrony metric $\langle\sigma^2\rangle_x$ (lines 785–787) outline the required content to replace the commented-out text. Below is a complete, publication-ready draft that fully describes the 3 panels of Fig. 4, correlates them with Floquet stability, addresses non-synchronized regimes, and transitions into the phase diagram.]

[**SUGGESTED DRAFT FOR Section V Discussion of Figure 4:**
We characterize synchronization with the external drive through the stroboscopic variance metric $\langle\sigma^2\rangle_x$, defined by sampling the displacement of the DW $u(x, t + 2T_{\rm drive}) - u(x, t)$ at intervals of twice the driving period $T_{\rm drive} = 2\pi/\Omega$. For each point on the wall, we compute the variance of these stroboscopic displacements over time, and then average across the length of the wall. In a phase-locked state, the stroboscopic displacement recurs identically in every cycle, yielding $\langle\sigma^2\rangle_x \approx 0$, whereas in unsynchronized regimes cycle-to-cycle fluctuations produce a finite value $\langle\sigma^2\rangle_x > 0$. Sampling at twice the period detects locking to both integer harmonics ($\langle\dot\phi\rangle = n\Omega$) and half-integer subharmonics ($\langle\dot\phi\rangle = n\Omega/2$).

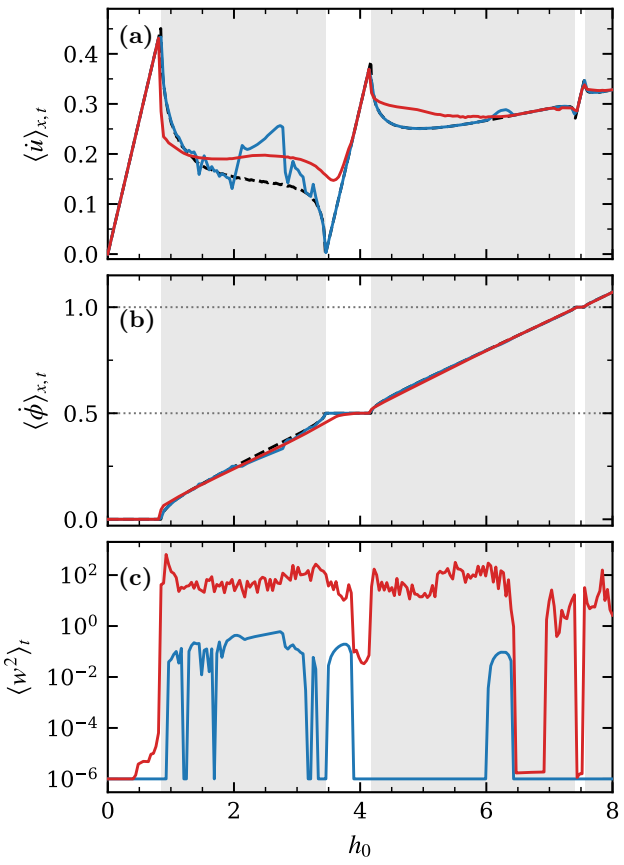


Figure 4. Flexible extended domain wall dynamics. (a) Mean DW velocity $\langle\dot u\rangle_{x,t}$ and (b) mean precessional velocity $\langle\dot\phi\rangle_{x,t}$ for extended walls of sizes $L/L_0 = 8$ and 1024, together with the rigid-wall result (dashed black lines). (c) Steady-state roughness $\langle w^2\rangle_t$ for the two extended-wall sizes. Values smaller than 10^{-6} are shown at 10^{-6} for visualization. White and gray regions correspond to synchronized and unsynchronized rigid-wall solutions, respectively. Horizontal dotted lines in (b) indicate the Shapiro steps at $\langle\dot\phi\rangle_{x,t} = \Omega/2 = 0.5$ and $\Omega = 1$. All simulations were performed with $\alpha = 0.27$, $\Omega = 1$, and $h_{\rm a} = 3$.

Figure 4 summarizes the steady-state response of extended walls for a representative AC amplitude $h_{\rm a} = 3$ as a function of the DC field h_0, comparing small ($L/L_0 = 8$) and large ($L/L_0 = 1024$) systems against the 1D rigid-wall prediction (dashed black curves). In Fig. 4(b), both extended walls display clear Shapiro synchronization steps in the mean precessional velocity $\langle\dot\phi\rangle_{x,t}$, locking precisely to the rigid-wall plateaus at $\langle\dot\phi\rangle_{x,t} = \Omega/2 = 0.5$ and $\Omega = 1$. Outside these steps (gray shaded intervals), the precessional velocity increases monotonically with h_0, exhibiting characteristic Walker-breakdown-like spikes near each unlocking transition.

The translational DW velocity $\langle\dot u\rangle_{x,t}$, shown in Fig. 4(a), exhibits marked differences between the two

system sizes. For the small wall ($L/L_0 = 8$), the velocity remains virtually indistinguishable from the rigid-wall curve over broad intervals of h_0. In the large system ($L/L_0 = 1024$), by contrast, substantial departures from the rigid-wall behavior occur near and outside the synchronization boundaries.

These dynamical differences are strongly correlated with the domain-wall roughness $\langle w^2 \rangle_t$ depicted in Fig. 4(c). While the small DW drops to the numerical floor ($\langle w^2 \rangle_t \leq 10^{-6}$, representing a macroscopically flat interface) across wide parameter windows, the large system develops pronounced roughness ($10^{-3} \sim 10^{-1}$) over most of the driving range, punctuated only by narrow windows of reduced roughness.

Within the synchronization steps (white regions), these nonlinear behaviors can be understood directly through the linear Floquet stability analysis of Section IV. When all spatial modes of the flat synchronized state satisfy $|\mu_\kappa| \leq 1$ [cf. Fig. 3(b)], perturbations decay exponentially and the extended wall settles into a perfectly flat, synchronized configuration. Because the small system possesses a discrete mode spectrum with fundamental mode $\kappa_1 = 2\pi/8 \approx 0.785$, it easily circumvents narrow instability tongues that fall between its allowed modes, thereby remaining flat over much wider ranges than the large system. In the large system ($L/L_0 = 1024$), the high mode density ensures that unstable modes are populated whenever a band with $|\mu_\kappa| > 1$ opens, triggering spatial corrugation.

Crucially, in the unsynchronized parameter intervals (gray regions), where Floquet analysis cannot be defined, the nonlinear simulations show that the wall can either maintain spatial flatness (as observed in the small system for $h_0 > 3.8$) or develop turbulent roughening. Furthermore, within the synchronized steps, spatial corrugation can coexist with temporal phase locking without destroying synchrony. These findings reveal that temporal synchronization and spatial stability are independent, cooperating mechanisms. Their interplay gives rise to five distinct dynamical regimes, which we classify systematically in the full phase diagram below.]

[**REC:** Figure 5 Label Consistency: In the figure panel and caption, the non-precessional phase is now harmonized to "PF" (Pinned Flat) across the figure legend, bottom DC strip, and caption, fully consistent with Table I and the other two-letter acronyms (SF, SP, NF, NR).]

[**REC:** Figure 7 Caption Clean-up: (i) Remove the Spanish draft title "Propuesta 2 (cascada)". (ii) Update "SC regime" to "SP phase" to match Table I. (iii) Suggested clean caption: "Spatiotemporal profiles of a section of the domain wall in the steady-state synchronized periodic (SP) phase ($h_0 = 4$, $h_a = 3$, $\alpha = 0.27$, and $\Omega = 1$), displayed at evenly spaced intervals over one full driving period $T = 2\pi/\Omega$. The wall undergoes periodic transverse breathing and inversion without developing a turbulent cascade into higher spatial modes."]

[**REC:** Figure 8 Caption Harmonization: Change "SC phase" → "SP phase". Note that in panel (a), roughness saturates to a size-independent constant $\langle w^2 \rangle_t \approx 0.05$ as $L \to \infty$, demonstrating genuine long-range periodic order in sharp contrast to the self-affine divergence $\langle w^2 \rangle_t \propto L$ characteristic of the rough NR phase.]

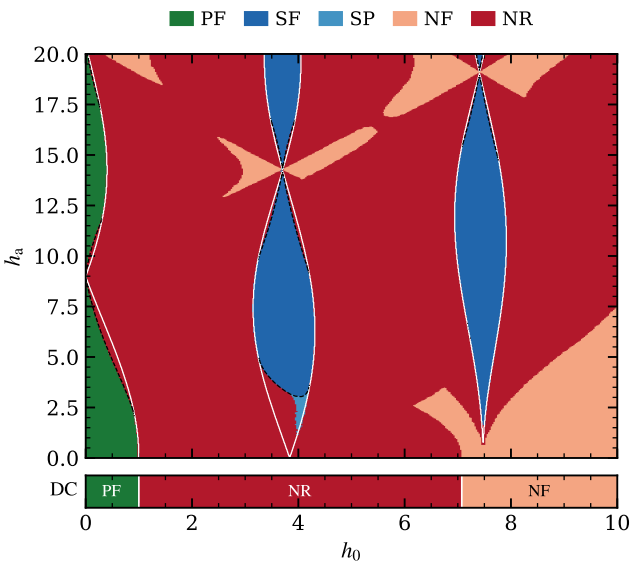


Figure 5. Dynamical phase diagram for an extended domain wall in the large-system limit ($L/L_0 = 256$), showing the five dynamical regimes: pinned flat (PF), synchronized flat (SF), synchronized periodic (SP), non-synchronized flat (NF), and non-synchronized rough (NR), for $\alpha = 0.27$ and $\Omega = 1$. Continuous white and dashed black curves represent the boundaries of synchrony and linear stability ($\mu = 1$) of the corresponding rigid flat wall, respectively. The horizontal strip at the bottom displays the purely DC-driven phases ($h_a = 0$): PF for $0 \leq h_0 \leq 1$, precessional rough (NR) for $1 < h_0 < h_c$, and precessional flat (NF) for $h_0 \geq h_c$, with $h_c \approx 7.08$. The small-system phase diagram ($L/L_0 = 8$) is presented separately in Appendix C.

VI. DYNAMICAL PHASES

We are now in a position to discuss the main result of this work: the full dynamical phase diagram in the (h_0, h_a) plane at fixed AC frequency. Figure 5 shows five distinct dynamical regimes: pinned flat (PF), synchronized flat (SF), synchronized periodic (SP), non-synchronized flat (NF), and non-synchronized rough (NR). The phase diagram corresponds to $L/L_0 = 1000$, for which finite-size effects are negligible on the scale considered. Finite-size effects become visible for substantially smaller systems (see Appendix C), so we use the $L/L_0 = 1000$ results as representative of the $L \to \infty$ dynamical phase diagram. Synchronization is identified by $\sigma < 0.01$. The continuous and dotted gray lines indicate, respectively, the synchronization boundaries and the linear-stability boundaries of the corresponding syn-

Phase	u spatial structure	ϕ synchronization	DW average precession
PF	Flat	Pinned	Absent
SF	Flat	Synchronized	Coherent
SP	Periodic (controlled by Floquet instabilities)	Synchronized	Coherent
NF	Flat	Non-synchronized	Coherent
NR	Rough (Edwards-Wilkinson)	Non-synchronized	Incoherent / chaotic

Table I. Characterization of the dynamical phases in terms of the spatial structure of u, synchronization of the domain-wall precession ϕ with the external drive, and coherence of the precessional dynamics. The SF and SP phases are synchronized and exhibit coherent domain-wall precession, while the NF phase corresponds to coherent but non-synchronized precession. In the NR phase, the precession becomes incoherent and chaotic, accompanied by Edwards–Wilkinson roughening of u in the thermodynamic limit. In the SP phase, the periodic spatial structure is characterized by a wavelength selected by the narrow band of linearly unstable modes.

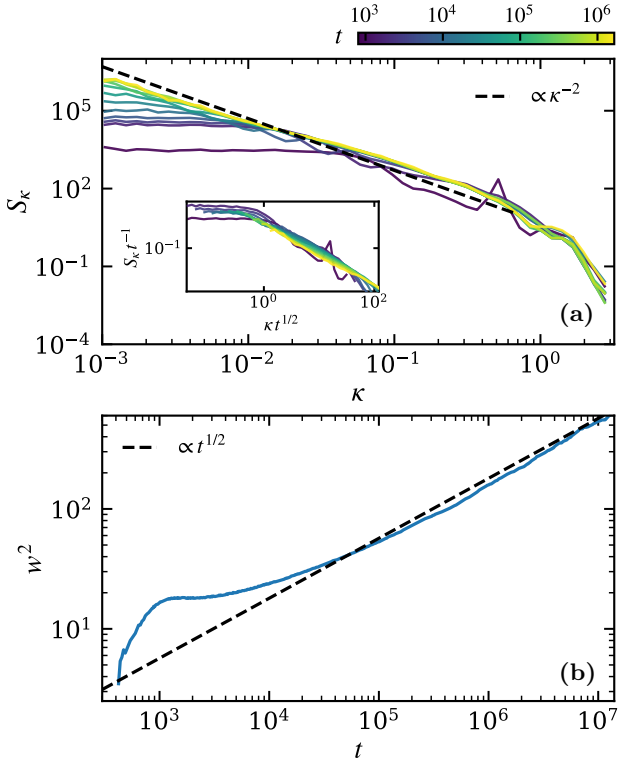


Figure 6. Universal scaling and exponent analysis of the non-synchronized rough (NR) phase ($h_0 = 5$, $h_a = 3$, $\Omega = 1$, $\alpha = 0.27$, for $L/L_0 = 1000$). (a) Main panel: Structure factor S_κ as a function of wavenumber κ at different times t during the transient growth regime, with the dashed line indicating the asymptotic power-law decay $S_\kappa \propto \kappa^{-2}$. Curves are color-coded by time t according to the colorbar above. Inset: Scaling collapse onto the universal master curve $S_\kappa t^{-1}$ versus $\kappa t^{1/2}$, demonstrating dynamic scaling consistent with $S(\kappa, t) \sim \kappa^{-(1+2\zeta)} F(\kappa t^{1/z})$ with roughness exponent $\zeta = 1/2$ and dynamic exponent $z = 2$. (b) Domain-wall roughness w^2 as a function of time t; the dashed line illustrates the characteristic growth regime $w^2 \propto t^{1/2}$. These scaling relations confirm that the NR phase under combined DC+AC driving belongs to the one-dimensional Edwards–Wilkinson universality class.

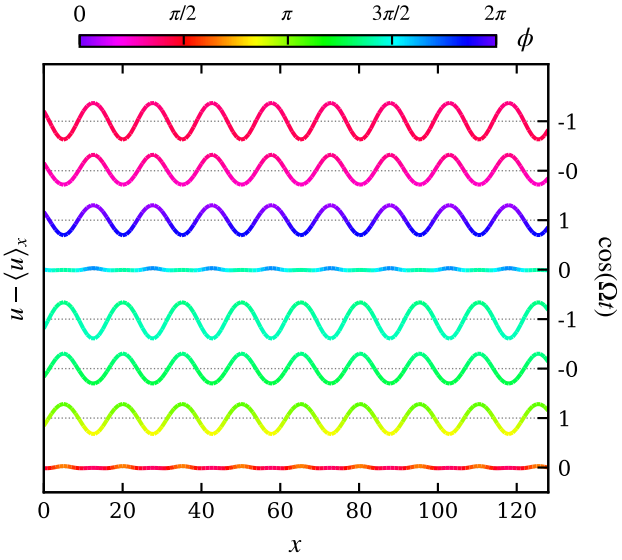


Figure 7. Propuesta 2 (cascada)
Snapshots of a section of the DW in the steady-state SC regime ($h_0 = 4$, $h_a = 3$, $\alpha = 0.27$ and $\Omega = 1$), taken at evenly spaced intervals over a full period of its motion.

chronized flat rigid-wall solutions. These boundaries provide a reference for the long-time dynamics of the full nonlinear extended-wall equations: the continuous lines delimit the parameter regions where the corresponding rigid-wall solutions are synchronized, while the dotted lines indicate where these synchronized flat solutions become spatially unstable. [**REC:** System Size Accuracy in Text: Change "$L/L_0 = 1000$" to "$L/L_0 = 256$" (in both sentences) to match Figure 5 and its caption where the simulated large system is explicitly $L/L_0 = 256$.]

In the PF phase, $\langle \dot{\phi} \rangle = 0$, so the DW does not exhibit average internal precession, although its mean position $\langle u \rangle$ can still evolve in time. The effective Walker breakdown field in presence of AC drive follows $h_W \approx |J_0(\alpha h_a/\Omega)|$, as predicted by the equality in Eq.(23) for the particular $n = 0$ case. The remaining four regimes combine temporal synchronization with distinct spatial structures. The SF and SP phases are synchronized,

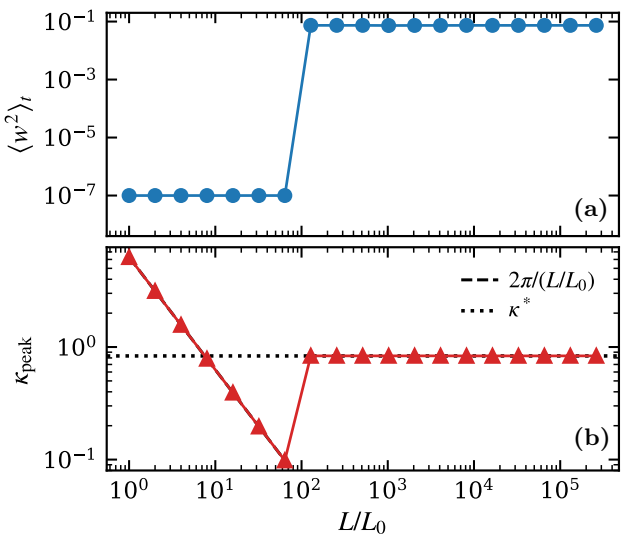


Figure 8. Finite size analysis of the SC phase. (a) DW rugosity $\langle w^2 \rangle_t$ and (b) wavenumber of the maximum peak in the structure factor $\kappa_{\rm peak}$ as a function of L/L_0. The long range spatiotemporal order is dominated by the most linearly unstable mode κ^* (dotted line) of the synchronized phase. For large enough systems, the roughness converges to a constant value independent of system size. For small sizes, the structure factor is dominated by the lowest available mode $2\pi/(L/L_0)$ (dashed line) and the DW is flat due to the lack of modes in the narrow instability band.

whereas NF and NR are non-synchronized. Spatially, SF and NF remain essentially flat, while SP develops a periodic corrugation and NR exhibits roughening. The SP regime occupies a comparatively narrow regions, for relatively low values of h_a, at the bottom of the synchronization regions. [**REC:** Typo: Change "occupies a comparatively narrow regions" → "occupies a comparatively narrow region".]

The synchronization regions of the extended domain wall remain close to those obtained from the rigid-wall dynamics (delimited by the continuous lines) over most of the phase diagram. These lines are described by Eqs.(23) and (24). Deviations due to the flexural modes of the extended DW are most pronounced near the edges of the synchronization steps and in the region where the SP state develops. In these regions, the Floquet stability analysis of the synchronized rigid-wall solution provides a direct indication of the onset of spatial instabilities (indicated by the dotted lines). In Fig. 5 we also observe that the dotted lines provide an approximation for the onset of instabilities inside the synchronized phase of the rigid DW. Thus, synchronization and spatial stability are related but distinct properties of the driven wall. [**REC:** Clarification of Dotted Stability Boundary (Addressing Fabricio's note): Clarify in the text that the dotted line marks the lower onset of linear stability for the synchronized flat state: the region below this dotted line is linearly unstable to flexural corrugations, explaining why the SP phase appears precisely in the unstable sub-tongue where synchrony holds but flat walls destabilize.]

The AC drive can qualitatively modify the dynamics observed under pure DC driving. It can suppress roughening and stabilize a synchronized flat state, or it can synchronize a spatially unstable state, leading to the SP regime. Conversely, non-synchronized flat states can also occur under AC driving. The resulting phase diagram therefore reflects the interplay between temporal synchronization and spatial stability rather than a single synchronization transition. Since the flat phases indistinguishable from the rigid-wall solutions, in the following we solely characterize the non-flat phases. [**REC:** Grammar Typo: Change "Since the flat phases indistinguishable" → "Since the flat phases are indistinguishable".]

We first characterize the NR regime. To quantify its spatial structure, we analyze the roughening process and consider the *time-dependent* structure factor starting from an initially perturbed flat DW

$$S_\kappa = \left\langle |\hat{u}_\kappa(t)|^2 \right\rangle , \tag{32}$$

and the *time-dependent* global roughness

$$w^2 = \left\langle u^2(t) \right\rangle_x - \left\langle u(t) \right\rangle_x^2 . \tag{33}$$

where the averages are done over random realizations of the perturbed initial DW.

During the transient growth regime that nucleates the NR phase, shown in Fig. 6, the structure factor follows the scaling form

$$S_\kappa \propto \kappa^{-2} F(\kappa\, l(t)), \tag{34}$$

with a growing length-scale $l(t) \sim t^{1/2}$, while $F(x) \sim x^2$ for small x and approaches a constant for large x. As shown in the inset of Fig. 6(a), the rescaled data $S_\kappa t^{-1}$ collapse onto a single universal curve when plotted against $\kappa t^{1/2}$. Correspondingly, the roughness grows as

$$w^2 \propto t^{1/2}. \tag{35}$$

These scaling properties are consistent with the one-dimensional Edwards–Wilkinson universality class, where $w^2 \sim t^{2\zeta/z}$ and $S(q) \sim q^{-(1+2\zeta)} F(qt^{1/z})$, with a roughness exponent $\zeta = 1/2$ and dynamic exponent $z = 2$. The NR phase under DC+AC drive is thus in the same the universality class as in the purely DC-driven case[10]. [**REC:** NR Universality & Effective Temperature (Addressing Ale's note): (i) Typo: "same the universality class" → "same universality class". (ii) Add: "Furthermore, the steady-state structure factor exhibits an asymptotic $S(\kappa) \approx T_{\rm eff}/\kappa^2$ divergence, reflecting a deterministically induced effective temperature $T_{\rm eff}$ originating from chaotic Bloch line nucleation and collisions. The height distribution $P(u)$ is Gaussian with vanishing skewness, ruling out Kardar–Parisi–Zhang (KPZ) universality and confirming linear Edwards–Wilkinson scaling under AC driving."]

We next turn to the SP regime. Figure 7 shows representative configurations over two periods of the AC drive. The wall develops a periodic corrugation whose amplitude oscillates in time, while the spatial profile recurs periodically. The roughness approaches a finite value as the system size increases, and the dominant wave number remains finite, indicating that the corrugation is not a finite-size artifact. [**REC:** Mechanism of Cascade Arrest in the SP Phase (Addressing Ale's note): Explain why unstable modes do not trigger spatiotemporal chaos: "Remarkably, despite possessing a finite band of linearly unstable modes, the SP phase does not degenerate into spatiotemporal chaos. Strong temporal phase-locking to the AC drive rigidly pins the phase of the dominant mode κ^*. This phase coherence arrests the multi-wave resonant interactions that otherwise drive an energy cascade toward higher and lower modes in the purely DC regime[10], trapping the system in a stable, periodically corrugated limit cycle."]

The spatial periodicity of the SP phase is directly related to the Floquet instability of the synchronized flat state. In the unstable region, a finite band of wave numbers has Floquet multipliers larger than unity, with a maximum growth rate at a characteristic wave number κ^*. The nonlinear SP state is dominated by a wave number κ_{peak} that approaches a finite value with increasing system size and is consistent with the Floquet-selected κ^* (indicated by the dashed horizontal line). Figure 8 shows that both the roughness and κ_{peak} converge as L increases. For sufficiently small systems, the allowed wave numbers may not sample the unstable band adequately; in that case the wall can remain effectively flat even when the corresponding continuum system is spatially unstable.

The finite-size analysis therefore demonstrates that the periodic corrugation in the SP regime has an intrinsic spatial scale selected by the linear instability of the synchronized state. In the large-system limit, the SP phase exhibits long-range spatial periodic order together with temporal synchronization.

The main properties of the five dynamical phases are summarized in Table I.

VII. CONCLUSIONS

We investigated synchronization in extended domain walls driven by combined DC and AC magnetic fields and its relation to spatial order. We showed that synchronization can occur even in very large systems exhibiting spatiotemporal chaos in the purely DC-driven regime. Synchronization is generally accompanied by a suppression of this chaotic dynamics and, in most cases, by a flattening of the domain wall. However, synchronized regimes with a periodically corrugated spatial structure also exist. In this regime, the corrugation is dominated by a finite band of linearly unstable modes with a well-defined maximum growth rate, linking the spatial structure of the nonlinear state to the Floquet instability of the synchronized flat state. The resulting dynamical phase diagram, shown in Figure 5, summarizes these different regimes.

The AC field amplitudes required to drive a chaotic domain wall into the SP phase are of the order of the Walker field. For a thin Co/Pt multilayer, the minimum AC amplitude required is of order 10 mT, with frequencies on the order of 100 MHz. These conditions are experimentally accessible (see Appendix A for the correspondence between the model parameters and physical units). The finite-size effects identified in this work may also be relevant for experimentally accessible micrometer-scale systems.

The framework developed here neglects disorder and thermal fluctuations[20,21], which may affect the experimental observation of the predicted dynamical regimes. Nevertheless, the present theory provides a basis for extending the analysis to more realistic domain-wall models that incorporate these effects. The framework can also be generalized to higher-dimensional geometries and to electrical current drives, providing directions for future studies.

Beyond these extensions, our results demonstrate that a spatially extended nonlinear system exhibiting deterministic spatiotemporal chaos can synchronize to an external periodic drive. We further show that synchronization can suppress the chaotic dynamics while preserving spatial structure, leading either to a synchronized flat state or to a synchronized periodically corrugated state with long-range spatiotemporal periodic order. This coexistence of temporal synchronization and spatial instability provides a distinctive dynamical regime of driven precessing domain walls. [**REC:** Conclusions Enhancement — Spintronic Outlook: Add an outlook paragraph on spin-orbit torques and device applications: "In addition to magnetic field driving, equivalent effective driving forces can be exerted through spin-orbit torques (SOT) in heavy-metal/ferromagnet bilayers (such as Pt/Co/AlOx). Radio-frequency AC currents in the gigahertz regime can deliver effective field amplitudes of tens of milliteslas, providing a versatile on-chip platform to observe these synchronization-induced transitions. Technologically, the ability to switch deterministically between flat (high mobility) and corrugated (low mobility) states under AC gating offers a promising mechanism for low-power spintronic microwave detectors, frequency filters, and physical reservoir computing."]

ACKNOWLEDGMENTS

We thank T. Giamarchi for discussions. Simulations used the Gerencia de Física HPC cluster at CAB-CNEA (SNCAD, Argentina).

Appendix A: Experimental conditions

We work out the conversion of our nondimensional model into physical units for the case of a standard Co/Pt multilayer, with parameters[18] $\alpha = 0.27$, $A_{\rm ex} = 1.4 \times 10^{-11}$ J/m, $M_{\rm s} = 9.1 \times 10^5$ A/m, $K_{\rm u} = 8.4 \times 10^5$ J/m^3 and $N_{\rm n} = 0.016$.

The length scales for this material are $\Delta \approx 6.6$ nm and $L_0 \approx 8.77\Delta \approx 58.0$ nm. This means that the values of L/L_0 of 7, 1000, and 10 000 in units of L_0 used throughout our work correspond to DWs of lengths 406 nm, 58.0 μm, and 580 μm, respectively. The limiting thermodynamic behavior begins to occur for lengths of order $100\,L_0 \approx 5.8$ μm. [**REC:** Physical Length Scales Update: Update the representative values of L/L_0 from obsolete values ($L/L_0 = 7, 1000, 10000$) to the actual simulation sizes used in the paper: $L/L_0 = 8$ (≈ 464 nm), $L/L_0 = 256$ (≈ 14.8 μm), and $L/L_0 = 1024$ (≈ 59.4 μm). Note that $L/L_0 = 1000$ corresponds to 58.0 μm, while the finite-size scaling in Fig. 8 extends up to $L/L_0 \approx 2.6 \times 10^5 \approx 15$ mm.]

The time unit in this system is $T_0 \approx 0.33$ ns, which corresponds to a frequency unit of 3.03 GHz. The angular frequency $\Omega = 1$ used in this work corresponds to a real frequency of 480 MHz. The wavelength corresponding to this frequency is 0.6 m, many orders of magnitude above both length scales of the system.

The Walker breakdown field is $H_{\rm W} \approx 2.48$ mT. As a specific example, to drive the system in the SF regime at $H_0 = 4H_{\rm W} \approx 10$ mT, with a frequency of $\Omega = 1$ (480 MHz), the AC field amplitude should be at least $H_{\rm a} \approx 3.5 H_{\rm W} \approx 9$ mT.

Appendix B: Numerical methods

All systems of ordinary differential equations (ODEs) were solved using a fourth-order Runge-Kutta method[22]. This includes the system of two coupled, non-autonomous nonlinear ODEs for the Walker model of rigid DWs, Eqs. (6) and (7), and the linear system of ODEs, Eq. (28), used to find the Floquet multipliers of the synchronized rigid wall solution in the extended DW model in Section III.

The partial differential equations (PDEs) for the full u–ϕ model (Eqs. (5)) were solved using a pseudo-spectral Crank-Nicolson scheme, following the method outlined by Pusiol et al.[10]. To do this, we first note that by defining the complex variable[23]

$$z(x,t) = u(x,t) - i\phi(x,t)\,, \tag{B1}$$

the u–ϕ model is equivalent to the single complex PDE

$$\dot{z} = \frac{\alpha - i}{2}[\alpha h(t) + i\sin(2\,\mathrm{Im}(z)) + z'']\,. \tag{B2}$$

Now we separate the right-hand side of Eq. (B2) into a linear and a nonlinear part:

$$\begin{aligned} \mathcal{L}(z) &= \frac{\alpha - i}{2} z''\,, \\ \mathcal{N}(z) &= \frac{\alpha - i}{2}[\alpha h(t) + i\sin(2\,\mathrm{Im}(z))]\,. \end{aligned} \tag{B3}$$

The linear term is stepped according to the Crank–Nicolson method, and the nonlinear term is stepped with an explicit Euler step:

$$\frac{z^{n+1} - z^n}{\delta t} = \frac{1}{2}\mathcal{L}(z^{n+1} + z^n) + \mathcal{N}(z^n)\,. \tag{B4}$$

By transforming to Fourier space, $\hat{\mathcal{L}}_k = -\frac{\alpha - i}{2}k^2$, and we can derive an explicit formula for the evolution of the k-th Fourier mode:

$$z_k^{n+1} = \frac{\left(1 + \frac{\delta t}{2}\hat{\mathcal{L}}_k\right) z_k^n + \delta t \widehat{\mathcal{N}(z^n)}_k}{\left(1 - \frac{\delta t}{2}\hat{\mathcal{L}}_k\right)}\,. \tag{B5}$$

At each step, the nonlinear term is calculated in real space and then Fourier transformed. This requires tracking z in both real and Fourier space.

We implemented this method in C++ and CUDA C++, developing an implementation-agnostic interface that allows us to run identical simulations on either a CPU or a GPU, as needed. The GPU implementation outperformed the CPU for system sizes requiring about 1 000 samples of z, and was essential in order to study system sizes bigger than $L/L_0 = 1000$.

We chose the spatial resolution such that macroscopic properties are stable and the smallest features are resolved. Then, we picked a time resolution dt small enough that results are independent of small variations.

[**REC:** Appendix B Numerical Details (Addressing Ale's note): Provide concrete numerical parameters to complete this section: "In our simulations, we used a spatial discretization step $\Delta x = 0.5\,L_0$ (corresponding to ≈ 29 nm in physical units, well below the characteristic Bloch line width), discretizing a system of size $L/L_0 = 1000$ into $N = 2048$ grid points (or $N = 16384$ for $L/L_0 = 10\,000$). The integration time step was fixed to $\delta t = 0.005\,T_0 \approx 1.65$ ps (corresponding to ≈ 1256 steps per drive cycle at $\Omega = 1$), which ensures numerical stability and prevents aliasing in the explicit nonlinear step. Total integration times ranged from $t_{\rm max} = 2000\,T_0$ to $5000\,T_0$ (≈ 0.66–$1.65\,\mu$s), discarding initial transients of at least $1000\,T_0$ before computing time averages. Numerical convergence was verified by halving the time step ($\delta t = 0.0025\,T_0$) and doubling the spatial resolution ($\Delta x = 0.25\,L_0$), confirming that steady-state velocities, roughness, and structure factor peak locations remain invariant to within numerical precision."]

The source code for the simulator is freely available in Ref. 24.

Appendix C: Finite Size Effects

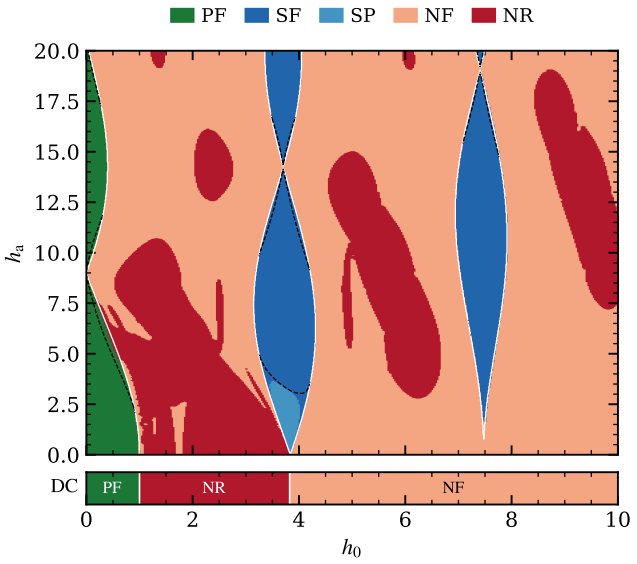


Figure 9. Dynamical phase diagram for a small extended domain wall of size $L/L_0 = 8$, driven by combined DC and AC magnetic fields with $\alpha = 0.27$ and $\Omega = 1$. The five dynamical regimes are: pinned flat (PF), synchronized flat (SF), synchronized periodic (SP), non-synchronized flat (NF), and non-synchronized rough (NR). Continuous white and dashed black curves indicate the rigid-wall synchrony and linear-stability boundaries, respectively. The horizontal strip at the bottom corresponds to purely DC driving ($h_a = 0$), showing the non-precessional regime (PF, $0 \le h_0 \le 1$), the precessional rough regime (NR, $1 < h_0 < h_c$), and the precessional flat regime (NF, $h_0 \ge h_c$), where $h_c \approx 3.8$ for $L/L_0 = 8$ due to the discrete mode spectrum.

In Figure 9, we show the dynamical phase diagram for a relatively small system, $L/L_0 = 8$. The differences with the large-system phase diagram (Figure 5) highlight the pronounced role of finite-size effects. In particular, for the small system, the discrete set of allowed spatial modes $\kappa_n = 2\pi n/(L/L_0) \approx 0.785n$ limits the availability of modes that can fall within the linear instability band of the precessing DW. Under purely DC driving ($h_a = 0$), shown in the horizontal strip below the diagram, the first flexural mode $\kappa_1 \approx 0.785$ is only unstable within a restricted window $1 < h_0 < h_c$, where $h_c \approx 3.8$. For $h_0 > h_c$, all available discrete modes become linearly stable, so the domain wall precesses as a flat interface (NF phase), in sharp contrast with the thermodynamic limit ($L \to \infty$), where the dense spectrum of long-wavelength modes sustains the chaotic NR regime up to $h_c \approx 7$. Similarly, under AC driving, the SF and NF phases dominate a broader region of the phase diagram in the small system, while the chaotic NR regime is significantly reduced compared to large systems.

Appendix D: Second-order perturbation theory for rigid DW synchrony

We can continue the expansion of Eq. (15) to second order in ϵ. To do this we expand $\epsilon \sin(\psi_1) \approx \epsilon \sin\left(\psi^{(0)}\right) + \epsilon^2 \psi^{(1)} \cos\left(\psi^{(0)}\right)$, substitute into Eq. (13) and collect the terms of order ϵ^2. We obtain:

$$\dot{\psi}_1^{(2)} = \tilde{h}_0^{(2)} - \psi^{(1)} \cos\left(\psi^{(0)}\right). \tag{D1}$$

To find $\psi^{(1)}$, we integrate Eq. (18) using the Jacobi-Anger expansion for $\sin\left(\psi^{(0)}\right)$ and substituting $\tilde{h}_0^{(1)}$ from Eq. (21). This yields:

$$\psi^{(1)}(\tau) = \sum_{m \neq -n} \frac{J_m(\tilde{h}_a)}{n+m} \cos[(n+m)\tau + \psi_c]. \tag{D2}$$

In order for the second-order solution to remain bounded and 2π-periodic, we require again that the time average of Eq. (D1) vanishes: $\tilde{h}_0^{(2)} = \left\langle \psi^{(1)} \cos\left(\psi^{(0)}\right) \right\rangle_\tau$. To compute this average, we expand $\cos\left(\psi^{(0)}\right)$ analogously to the sine term:

$$\cos\left(\psi^{(0)}\right) = \sum_{k=-\infty}^{\infty} J_k(\tilde{h}_a) \cos[(n+k)\tau + \psi_c]. \tag{D3}$$

In the product $\psi^{(1)} \cos\left(\psi^{(0)}\right)$, the time average of terms of the form $\langle \cos(p\tau + \psi_c) \cos(q\tau + \psi_c) \rangle_\tau$ is nonzero only when $p = \pm q$. Filtering these resonant terms leads to two contributions to the second-order correction:

$$\tilde{h}_0^{(2)} = A + B \cos(2\psi_c), \tag{D4}$$

where, using the identity $J_{-v}(x) = (-1)^v J_v(x)$, we define the summation constants A and B as:

$$A = \frac{1}{2} \sum_{m \neq -n} \frac{J_m^2(\tilde{h}_a)}{n+m}, \tag{D5}$$

$$B = \frac{1}{2} \sum_{m \neq -n} (-1)^m \frac{J_m(\tilde{h}_a) J_{2n+m}(\tilde{h}_a)}{n+m}. \tag{D6}$$

However, shifting the index in our expression for B to $r = n + m$, we can rewrite to obtain:

$$B = \frac{1}{2} \sum_{r \neq 0} \frac{J_{r-n}(\tilde{h}_a) J_{-r-n}(\tilde{h}_a)}{r} = 0, \tag{D7}$$

since the sum terms are odd in r. This simplifies Eq. (D4) to $\tilde{h}_0^{(2)} = A$. Also note that $A = 0$ if $n = 0$, for similar reasons. The second order corrections vanish in the zeroth synchrony step.

Substituting in Eq. (D4), the DC field in the synchrony condition becomes:

$$\tilde{h}_0 \approx n + \epsilon^2 A + \epsilon (-1)^n J_n(\tilde{h}_a) \sin(\psi_c), \tag{D8}$$

or, in the original physical variables,

$$h_0 \approx \frac{n\Omega}{\alpha} + \frac{\alpha}{\Omega}A + (-1)^n J_n(\frac{\alpha}{\Omega}h_a)\sin(\psi_c). \quad \text{(D9)}$$

This is the same as the first-order case, but with a rigid, field-dependent offset in the center of the synchronization step $\frac{\alpha}{\Omega}A$. Consequently, the only second-order effect is a rigid shift in the synchronization steps, depending only on $\alpha h_a/\Omega$, while maintaining their widths.

In the limit $h_a = 0$, the only term contributing to A is the one proportional to J_0. For $n \neq 0$, $A = \frac{1}{2n}$: the shift is towards higher absolute DC fields and decreases with the absolute value of the step index. In the limit of $h_a \to \infty$, $A \to 0$ due to the decay of the Bessel functions, implying that the second order corrections are most significant for small AC fields.

Figure 2 compares first and second order approximations for the Shapiro steps location.

[**REC:** Appendix E / Supporting Information: Uncomment "\input{supporting_information}" to include the analytical secular Hill expansion as Appendix E. Update the section title in supporting_information.tex to remove Fabricio's note and read: "\section{Multimode secular Hill expansion for domain wall Floquet stability}". This provides the exact mathematical underpinning for the reentrant stability windows observed in Fig. 3.]

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